Spot–Vol Dynamics: SSR and Smile Transport

Moving the smile when spot moves, without recalibration

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Abstract

When the underlying moves, the implied-volatility smile moves with it, and a desk must know *how* — to risk-manage, to scenario, and to refresh a surface between calibrations. This note documents the Vol-Fitter's spot–vol dynamics. The organizing quantity is the skew-stickiness ratio $\mathrm{SSR} = (\mathrm{d}\sigma_{\mathrm{atm}}/\mathrm{d}\ln F)/\mathrm{skew}$, the fraction of the skew realized as an ATM-vol change per unit log-spot move. Three canonical regimes fix it: sticky-moneyness ($\mathrm{SSR} = 0$, the smile rides with the forward), sticky-strike ($\mathrm{SSR} = 1$, vol fixed at each strike), and sticky-local-vol ($\mathrm{SSR} \approx 2$ for short-dated skews). We give the analytic transport that realizes any SSR in one line, derive its ATM response, and then the *exact* sticky-local-vol transport (a Hagan-style log-moneyness displacement) and the local-vol-grid relabeling that implements it by simply re-indexing the surface's strike nodes. All of it is recalibration-free: a spot move refreshes the surface by arithmetic, not by re-solving. A worked example transports an equity smile under a -5% move in the three regimes.

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1 The question

A calibrated surface is a snapshot at one spot. The next tick moves spot, and the desk needs the new smile *now* — faster than a recalibration, and consistent with a chosen dynamics assumption. The assumptions differ in how much the ATM vol follows the skew as spot moves, and the summary statistic is the *skew-stickiness ratio*

$$\mathrm{SSR} = \frac{\mathrm{d}\sigma_{\mathrm{atm}}/\mathrm{d}\ln F}{s_0}, \qquad s_0 = \left.\frac{\partial \sigma}{\partial k}\right|_{k=0} \text{ (the ATM skew).} \tag{1}$$

Heuristic

Read SSR as "how much of the skew is real". The skew s_0 says that lower strikes have higher vol. When spot drops, does the ATM vol rise to where the old lower-strike vol was (SSR$=1$, sticky-strike), not move at all in moneyness terms (SSR$=0$, sticky-moneyness), or *overshoot* (SSR≈ 2, sticky-local-vol, the empirically common short-dated regime)? SSR interpolates these, and the transport below realizes any value of it.

2 The analytic transport

Let $h = \delta = \ln(F_{\mathrm{new}}/F_{\mathrm{old}})$ be the log-forward move and $\sigma_{\mathrm{old}}(k)$ the pre-move smile in log-moneyness. The Vol-Fitter transports the smile to new-forward moneyness k by

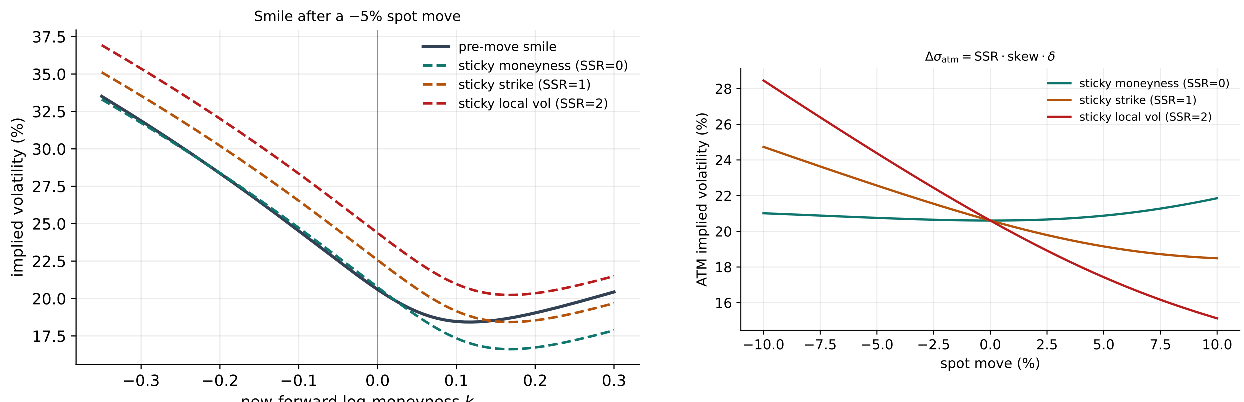
$$\boxed{\sigma_{\mathrm{new}}(k) = \sigma_{\mathrm{old}}(k+\delta) + (\mathrm{SSR}-1)\, s_0\, \delta.} \tag{2}$$

The first term is a *sticky-strike re-indexing* (a strike that was at moneyness $k+\delta$ before the move is at moneyness k after it); the second is a uniform *level adjustment* that dials in the desired ATM response while preserving the smile shape.

Proposition 1 (ATM response)**.** *Under (2), to first order in δ the ATM vol moves by $\Delta\sigma_{atm} = \mathrm{SSR}\cdot s_0 \cdot \delta$, consistent with the SSR definition (1).*

Proof. At $k=0$, $\sigma_{\mathrm{new}}(0) = \sigma_{\mathrm{old}}(\delta) + (\mathrm{SSR}-1)s_0\delta = \sigma_{\mathrm{old}}(0) + s_0\delta + (\mathrm{SSR}-1)s_0\delta + O(\delta^2) = \sigma_{\mathrm{old}}(0) + \mathrm{SSR}\, s_0\delta + O(\delta^2)$. □

So $\mathrm{SSR}=0$ leaves the smile a pure function of moneyness (sticky-moneyness: $\Delta\sigma_{\mathrm{atm}} = 0$), $\mathrm{SSR}=1$ cancels the level term and freezes vol at each strike (sticky-strike), and $\mathrm{SSR}=2$ doubles the skew's contribution. Figure 1 transports an equity smile (ATM skew $s_0 = -0.353$) under a -5% move in all three regimes; section 2 shows the linear ATM responses of slope $\mathrm{SSR}\, s_0$.



(a) transported smiles

(b) ATM response vs spot move

Figure 1: Transporting an equity smile under a -5% spot move in the three canonical regimes. The ATM response is linear with slope $\mathrm{SSR} \cdot s_0$.

3 Exact sticky-local-vol

The shift (2) is a first-order, shape-preserving approximation. For the *sticky-local-vol* regime there is an *exact* transport, because holding the local-volatility function fixed in absolute strike determines the new implied surface completely. In total-variance form, the horizontal-SSR transport is

$$\widetilde{w}_1^R(T, k) = w_0(T,\ k + R\, h_T), \tag{3}$$

and the exact sticky-local-vol transport replaces the linear displacement by the Hagan log-moneyness map

$$w_1^{\mathrm{LV}}(T, k) = w_0(T,\ \ell_T(k, h)), \qquad \ell_T(k, h) = \log\big(e^h(e^k + 1) - 1\big), \tag{4}$$

which is the exact relabeling of moneyness when the local-vol surface is held fixed in absolute strike and the forward moves by h. An optional ATM re-anchor $\sigma_{\mathrm{atm}} \to \sigma_0 + R\, \kappa_T h_T$ makes the linear SSR response exact even for large moves.

3.1 The local-vol-grid relabeling

When the dynamics are driven by the extracted local-vol *grid* (Note 04), the transport is even simpler: it is a *relabeling of the grid's strike nodes.* A grid node at absolute strike $K_i^0(t)$ moves to

$$K_i^{1,R}(t) = K_i^0(t)\, e^{(1-R/2)h_t}, \qquad \Longleftrightarrow \qquad x_i^1 = x_i^0 - \tfrac{R}{2}\, h_t, \tag{5}$$

in forward-normalized moneyness x. For $R = 2$ (sticky-local-vol) the factor is $e^0 = 1$: the grid is held fixed in absolute strike and re-priced through the Dupire PDE, which is the *exact* dynamics — here SSR is an *output* of the reprice, not an input, and the value 2 is only its short-maturity limit. For $R = 0$ (sticky-moneyness) the grid scales with the forward; $R = 1$ (sticky-strike) sits halfway.

This is why the local-vol regime is labeled both `sticky_local_vol` (the analytic Hagan transport) and `sticky_local_vol_grid` (the exact PDE reprice).

Heuristic

The grid relabeling (5) is the cleanest statement of the whole note: a stickiness regime is just a rule for *how the local-vol grid's strikes follow the spot.* $1 - R/2$ is the fraction of the spot move the grid follows — none of it ($R = 2$, grid frozen in strike), all of it ($R = 0$, grid rides the forward), or half ($R = 1$). Everything else — the implied-vol shift (2), the ATM response, the Hagan map — is a consequence of that one rule.

4 What is original here

The SSR and the stickiness regimes are standard (Bergomi, Hagan); the value here is the *unified, recalibration-free* implementation. The single transport (2) realizes *any* SSR (named regime or custom number) by arithmetic on the existing curve; the exact sticky-local-vol path (4)–(5) ties the analytic shift, the Hagan relabeling, and the local-vol-grid reprice into one parameter R; and because the local-vol regime can be either the analytic transport or the exact PDE reprice, the desk can trade speed for exactness on one slider. No spot move ever triggers a re-solve.

A Hyperparameter atlas

Table 1: Spot–vol-dynamics hyperparameters.

Knob	Default	Role
`ssr`	2.0	SSR value or named regime (`sticky_moneyness` / `sticky_strike` / `sticky_local_vol` / `sticky_local_vol_grid`).
R regime map	$\{0, 1, 2, 2\}$	SSR of the four named regimes.
ATM re-anchor	off	Optional exact linear SSR for large moves ($\sigma_0 + R\kappa_T h_T$).

Performance. The analytic transport (2) and the grid relabeling (5) are $O(\text{grid})$ arithmetic — no optimization, no PDE — so a spot move refreshes the whole surface essentially for free. Only the exact sticky-local-vol-grid reprice costs a Dupire PDE solve, and is chosen explicitly when exactness is wanted over speed. A spot move bumps the per-ticker spot version, so only the moved name's derived grid is re-transported.

References

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[3] J. Gatheral. *The Volatility Surface*. Wiley, 2006.